PROJECT DESIGN PHASE
Solution Architecture

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

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1. Introduction

The **Solution Architecture** describes the overall structure and interaction between the major components of the Personalized Networking Assistant (Smart Connect AI). It illustrates how user requests flow through the system, how the backend processes information, how external services are integrated, and how data is stored.

The architecture follows a **three-tier client-server architecture**, providing modularity, scalability, and easy maintenance.

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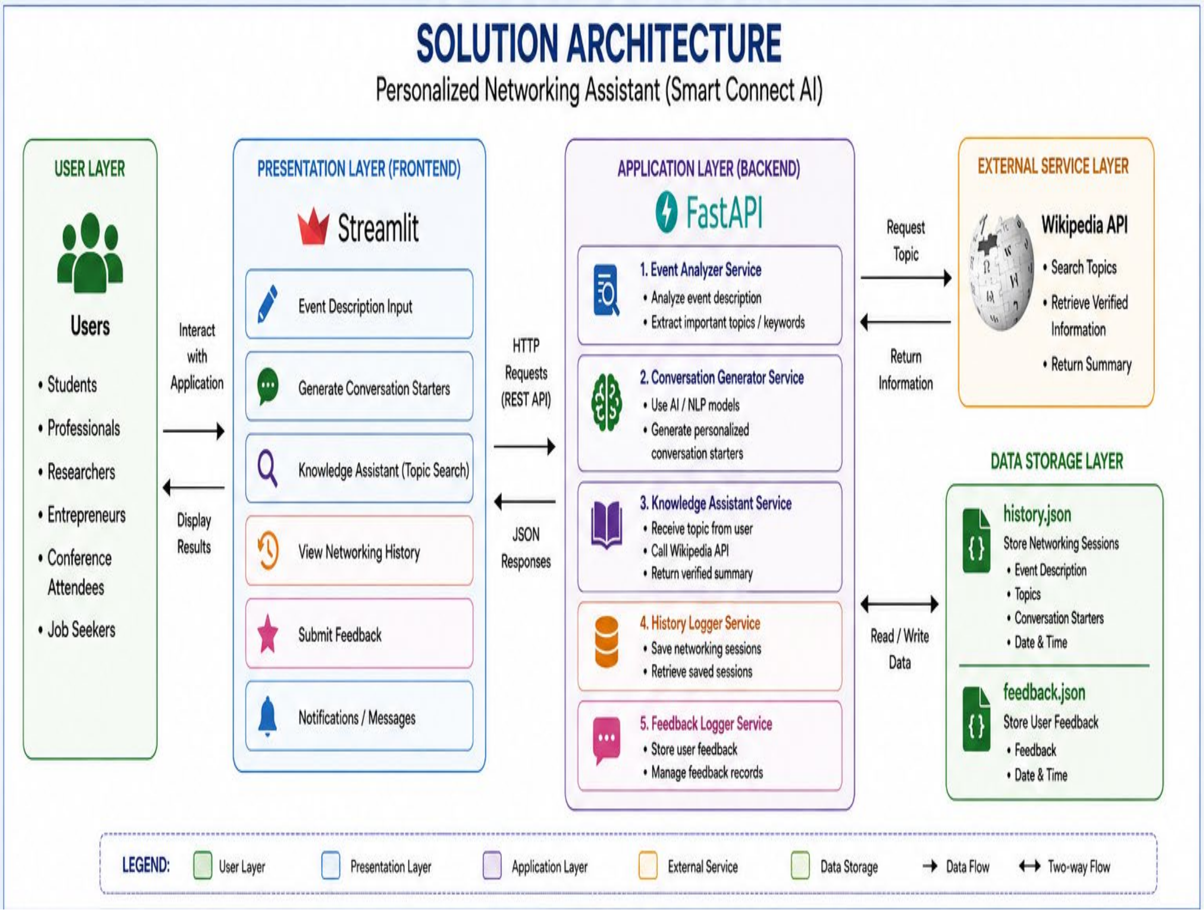
2. Solution Architecture Overview



3. User Layer

The User Layer represents users interacting with the application.

Users

- Students
- Professionals
- Researchers
- Entrepreneurs
- Conference Attendees
- Job Seekers

User Activities

- Enter event description
- Generate conversation starters
- Search networking topics
- Save networking sessions
- Submit feedback

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4. Presentation Layer (Frontend)

The Presentation Layer is developed using **Streamlit**.

Its responsibilities include:

- Displaying the homepage
- Collecting user inputs
- Displaying generated conversation starters
- Showing verified information
- Saving networking sessions
- Accepting user feedback
- Displaying notifications and messages

This layer communicates with the backend using HTTP REST APIs.

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5. Application Layer (Backend)

The backend is developed using **FastAPI**.

It consists of the following services.

Event Analyzer Service

Functions

- Analyze event descriptions
- Extract networking topics
- Process user input

Conversation Generator Service

Functions

- Generate intelligent conversation starters
- Produce personalized networking questions

Knowledge Assistant Service

Functions

- Receive user topic
- Connect with Wikipedia API
- Return verified information

History Logger Service

Functions

- Save networking sessions
- Retrieve saved sessions

Feedback Logger Service

Functions

- Store user feedback
- Manage feedback records

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6. External Service Layer

The system integrates the **Wikipedia API**.

Responsibilities include:

- Searching requested topics

- Retrieving verified summaries
- Providing trusted information
- Supporting the Knowledge Assistant module

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7. Data Storage Layer

The application stores data locally using JSON files.

history. json

Stores:

- Event Description
- Extracted Topics
- Generated Conversation Starters
- Date & Time

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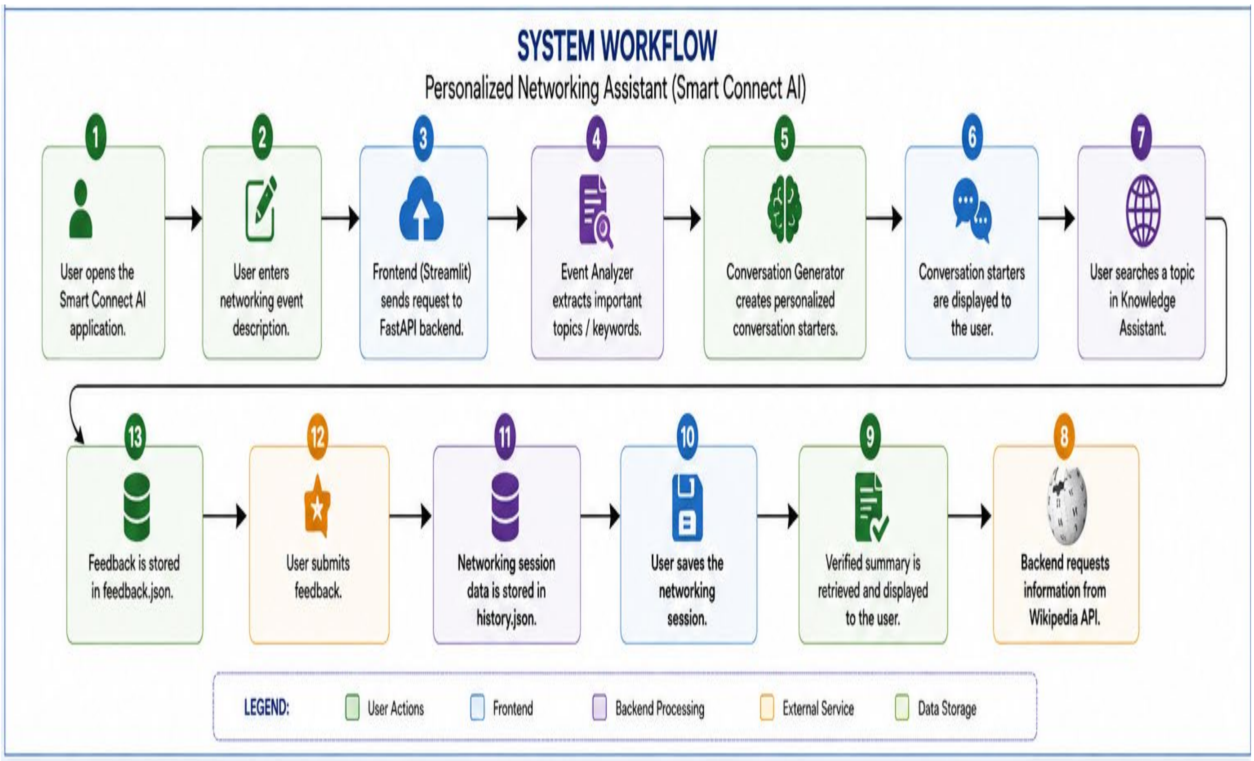
feedback. json

Stores:

- User Feedback
- Date & Time

8. System Workflow



9. Solution Architecture Components

Component	Technology	Purpose
User Interface	Streamlit	Collect user input and display output
Backend API	FastAPI	Process user requests
Event Analyzer	Python	Analyze event descriptions
Conversation Generator	Python (NLP)	Generate networking conversation starters
Knowledge Assistant	Wikipedia API	Retrieve verified information
History Logger	JSON	Save networking sessions
Feedback Logger	JSON	Store user feedback
HTTP Communication	Requests Library	Frontend-backend communication
Web Server	Uvicorn	Execute FastAPI application

10. Benefits of the Architecture

The proposed architecture offers several advantages:

- Modular design
- High scalability
- Easy maintenance
- Fast API communication
- User-friendly interface
- Reliable knowledge retrieval
- Local data storage
- AI-powered networking assistance
- Future extensibility
- Improved system performance

11. Future Scope

The architecture can be extended by integrating:

- OpenAI GPT models

- Firebase Authentication
- MongoDB Cloud Database
- LinkedIn API
- Voice Assistant
- Speech Recognition
- Recommendation System
- Mobile Application
- Multi-language Support

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12. Conclusion

The proposed **Solution Architecture** provides a robust, modular, and scalable framework for the **Personalized Networking Assistant (Smart Connect AI)**. By integrating **Streamlit**, **FastAPI**, **Python**, **Wikipedia API**, **Requests**, and **JSON-based storage**, the system delivers intelligent conversation generation, reliable knowledge verification, networking session management, and user feedback collection. This architecture ensures efficient communication between components while supporting future enhancements and advanced AI capabilities.